Diabetes Mellitus

Diabetes is a disease that affects how the body uses sugar. Sugar is a type of energy from food. People with diabetes have too much sugar in their blood. This can make them feel tired and thirsty.

Definition:

Diabetes is a disease where the body cannot control the sugar (glucose) in the blood.

This happens because the body does not make enough insulin or cannot use insulin properly.

Insulin is a hormone that helps sugar enter the cells for energy.

Types of Diabetes:

There are three main types:

- Type 1: The body stops making insulin because the immune system attacks the cells that produce insulin.
- Type 2: The body does not use insulin well and may not make enough insulin.
- Gestational Diabetes: Happens during pregnancy and usually goes away after birth.

Causes:

Type 1 diabetes is an autoimmune disease where the body's immune system attacks and destroys insulin-producing cells in the pancreas. As a result, the body makes little or no insulin. This type is usually diagnosed in children and young adults and requires daily insulin injections for survival. It is not caused by lifestyle factors.

Type 2 diabetes occurs when the body's cells do not use insulin properly (insulin resistance), and the pancreas does not produce enough insulin to maintain normal blood sugar levels. It is the most common type and is often linked to risk factors such as being overweight, lack of exercise, and family history. Lifestyle changes can delay or prevent Type 2 diabetes.

Gestational diabetes develops during pregnancy due to hormones that block insulin's action. It usually resolves after birth but increases the risk of developing Type 2 diabetes later.

Diabetes can be caused by genetics, lifestyle, or problems with the pancreas. Type 1 is caused by an autoimmune reaction. Type 2 is caused by insulin resistance, often linked to being overweight or inactive.

Risk Factors:

- Family history of diabetes
- Being overweight or obese
- Lack of exercise
- Older age
- Unhealthy eating
- Having high blood pressure or cholesterol

Clinical Picture

- Feeling very thirsty
- Going to the toilet often

- Feeling tired
- Losing weight without trying
- Cuts that heal slowly
- Blurred vision

Diagnosis:
Doctors check blood sugar levels with blood tests.

Common tests include

fasting blood sugar ≥ 126

2 hour postprandial blood glucose level ≥ 200mg/dl

HbA1c ≥ 6.5%, which shows average blood sugar over months.

Prevention:

- Eat healthy foods like fruits and vegetables
- Exercise regularly
- Maintain a healthy weight
- Avoid smoking

Complications:
If not treated, diabetes can cause heart problems, kidney damage, eye problems, and foot ulcers.

Acute complications:- include diabetic ketoacidosis (DKA), hypoglycaemia, hyperglycaemic hyperosmolar state (HHS) and hyperglycaemia. These complications can be fatal if not diagnosed early and managed appropriately

Chronic complications:-The long-term complications arise from prolonged high blood glucose levels and can be serious, sometimes leading to amputation or even blindness.

Cardiovascular Problems

Heart Attack and Stroke: High blood sugar, high blood pressure, and abnormal cholesterol levels increase the risk of atherosclerosis (narrowing of blood vessels), which can lead to heart attacks and strokes.

Nerve Damage

Diabetic Neuropathy: High glucose levels and reduced blood flow can damage nerves, leading to pain, tingling, burning, or loss of feeling in the hands and feet. Nerve damage can also affect autonomic functions, causing issues with digestion, blood pressure regulation, and erectile dysfunction.

Kidney Problems

Diabetic Nephropathy: Diabetes can damage the small blood vessels in the kidneys, impairing their ability to filter waste. This can progress to kidney disease and, in severe cases, end-stage kidney disease requiring dialysis or a transplant.

Eye Damage

Diabetic Retinopathy: High blood sugar can damage the blood vessels in the retina of the eye. This can lead to vision problems and even blindness if left untreated.

Foot Problems

Ulcers and Infections: Nerve damage and poor blood flow to the feet increase the risk of injuries, ulcers, and infections. Severe foot problems may sometimes require amputation.

Other Complications

Skin and Mouth Problems: People with diabetes are more prone to skin conditions and gum disease.

Infections: A weakened immune system can make people with diabetes more susceptible to common infections.

Sexual and Bladder Problems: Nerve damage can affect sexual function and bladder control.

Liver Disease: Diabetes can contribute to fatty liver disease and liver fibrosis, which can lead to cirrhosis.

Treatment:

- Type 1 needs insulin injections
- Type 2 can be treated with lifestyle changes, medicine, and sometimes insulin
- Regular monitoring of blood sugar is important
- Healthy eating and exercise help all types

In summary:

Feature	Type 1 Diabetes	Type 2 Diabetes	Gestational Diabetes
Cause	Autoimmune destruction of insulin cells	Insulin resistance and inadequate insulin	Insulin-blocking hormones during pregnancy
Age of Onset	Usually childhood or young adulthood	Usually adults, but can occur at any age	During pregnancy
Insulin Production	Little or none	Insulin produced, but inadequate or resisted	Variable, often normal after pregnancy
Treatment	Daily insulin injections	Lifestyle changes, oral medications, insulin	Lifestyle changes, sometimes insulin
Prevention	Not preventable	Can be delayed or prevented by healthy lifestyle	Usually not preventable, manage during pregnancy
Risk Factors	Genetic, environmental unknown	Obesity, inactivity, family history	Pregnancy-related factors

This comparison provides a clear differentiation between the main diabetes types for better understanding [10][11][12].

Which tests diagnose each type and their threshold values

For diagnosing diabetes types, the main tests and their threshold values are as follows:

1. **Fasting Blood Sugar (FBS) Test**
   - Diabetes is diagnosed if fasting blood sugar is **126 mg/dL (7.0 mmol/L)** or higher on two separate occasions.
   - Prediabetes is diagnosed if fasting glucose is **100-125 mg/dL (5.6-6.9 mmol/L)**.
2. **HbA1c Test (Glycated Hemoglobin)**
   - Diabetes: HbA1c level of **6.5% or higher**.
   - Prediabetes: HbA1c between **5.7% and 6.4%**.
   - HbA1c reflects average blood sugar over the past 2-3 months.
3. **Oral Glucose Tolerance Test (OGTT)**
   - Diabetes diagnosed if blood glucose is **200 mg/dL (11.1 mmol/L)** or higher 2 hours after drinking a glucose solution.
   - Prediabetes if 2-hour glucose is **140-199 mg/dL (7.8-11.0 mmol/L)**.
4. **Random Blood Sugar Test** (used if symptoms are present)
   - Diabetes if blood sugar is **200 mg/dL (11.1 mmol/L)** or higher anytime.

These tests apply broadly to diagnosing both Type 1 and Type 2 diabetes. Type 1 is usually diagnosed with symptoms plus elevated blood sugar or specific antibody tests, while Type 2 diagnosis relies on these blood sugar thresholds and clinical presentation. Gestational diabetes uses slightly different OGTT criteria during pregnancy (e.g., glucose levels at fasting, 1-hour, and 2-hour timings) per clinical guidelines.

Summary Table:

Test	Diabetes Diagnosis Threshold	Prediabetes Threshold
Fasting Blood Sugar	≥ 126 mg/dL (7.0 mmol/L)	100-125 mg/dL (5.6-6.9 mmol/L)
HbA1c	≥ 6.5%	5.7% - 6.4%
2-Hour OGTT	≥ 200 mg/dL (11.1 mmol/L)	140-199 mg/dL (7.8-11.0 mmol/L)
Random Blood Sugar	≥ 200 mg/dL (11.1 mmol/L)	Not typically used

For Gestational Diabetes, the OGTT criteria differ but generally involve:

- Fasting glucose ≥ 92 mg/dL (5.1 mmol/L)
- 1-hour glucose ≥ 180 mg/dL (10.0 mmol/L)
- 2-hour glucose ≥ 153 mg/dL (8.5 mmol/L)

These thresholds help guide diagnosis and treatment plans for diabetes [20][21][22].

What treatments and long-term management differ by type

The treatments and long-term management of diabetes differ by type primarily based on the body's insulin production and usage:

Type 1 Diabetes:

- Treatment requires daily insulin injections or an insulin pump since the body produces little or no insulin.
- Blood sugar levels must be regularly monitored.
- Long-term management focuses on balancing insulin doses with food intake and physical activity.
- Healthy lifestyle habits help but cannot replace insulin therapy.

Type 2 Diabetes:

- Initially managed with lifestyle changes such as healthy diet, weight loss, and regular exercise.
- Oral medications are commonly prescribed if lifestyle changes alone aren't enough.
- Insulin therapy may be needed if blood sugar remains uncontrolled.
- Long-term management includes monitoring blood sugar, blood pressure, and cholesterol to prevent complications.

Gestational Diabetes:

- Managed with healthy eating and exercise.
- Sometimes insulin is required if blood sugar levels are too high.
- Monitoring during pregnancy is essential.
- Usually resolves after delivery but requires follow-up to monitor for development of Type 2 diabetes.

Overall, Type 1 needs lifelong insulin treatment and careful blood sugar monitoring, whereas Type 2 treatment begins with lifestyle changes and medication, possibly progressing to insulin. Prevention and management of complications are critical parts of long-term care in all types, requiring regular medical check-ups and self-care routines.

The emphasis in long-term management is on maintaining blood sugar within target ranges to reduce risks of heart disease, kidney damage, vision loss, and nerve problems. Collaborative care with healthcare providers supports ongoing adjustment of treatment plans.

This approach highlights the differing medical needs and daily management responsibilities between diabetes types .

Pharmacologic Therapy

Early initiation of pharmacologic therapy is associated with improved glycemic control and reduced long-term complications in type 2 diabetes. Drug classes used for the treatment of type 2 diabetes include the following:

Biguanides

GLP-1 agonists

Selective sodium-glucose transporter-2 (SGLT-2) inhibitors

Dipeptidyl peptidase IV (DPP-4) inhibitors

Sulfonylureas

Alpha-glucosidase inhibitors

Thiazolidinediones (TZDs)

Insulins